

Three Insights Toward a Theory of Compositional Generalization in Learned Systems

Abstract

This note summarizes three connected insights toward a unified account of representation, generalization, and inductive bias in learned systems. The first frames data as a *holonarchic* structure whose elements have a *dispositional* character, operationalized through conditional Kolmogorov complexity expressed as a function of representational radius. The second derives from this a procedure for *mapping the frontier of known observations* and *estimating the location of novel ones* via threshold-based reconstruction and recursive association decomposition. The third is a phenomenological critique of the smoothness priors that dominate current regularization practice, arguing that they may reflect anthropic biases about cause and effect rather than properties of the data manifold itself.

1 Holonarchy and Dispositional K-Complexity

A data distribution D is treated not as a flat set of points but as a *holonarchic* structure: a hierarchy in which each element is simultaneously a whole (with internal compositional structure) and a part (entering as a component of larger wholes). Under this view, an observation x is not adequately characterized by its features alone — it is characterized by its *dispositions*: the set of compositional roles it can play across levels of the hierarchy.

We make this precise through conditional Kolmogorov complexity. Define

$K(x \mid D, r)$ = minimal description length of x given the subset of D within representational radius r of x .

The function $r \mapsto K(x \mid D, r)$ is the *dispositional curve* of x . Its shape encodes how the explanatory burden of x shifts as the conditioning neighbourhood grows. Three regimes are notable:

- For small r , $K(x \mid D, r)$ is large: the local neighbourhood is too small to compress x effectively.
- For intermediate r , $K(x \mid D, r)$ drops sharply at the radius at which x 's compositional structure becomes specifiable in terms of nearby holons.
- For large r , $K(x \mid D, r)$ asymptotes to a floor set by the prior complexity of the full data distribution.

The position of the inflection encodes the level of the holonarchy at which x is most naturally described. The disposition of x is the integral of associations that become available as r grows past this inflection.

2 Frontier Mapping and Generative Estimation via Two Thresholds

The dispositional view suggests an operational procedure for characterising any observation: identify two critical thresholds in a noise- or perturbation-conditioned reconstruction process.

Let $\tilde{x}_t$ be a perturbed version of x at noise level $t \in [0, T]$, and let $\hat{x}_t$ denote the reconstruction by a model conditioned on D . Define:

- $T_1(x)$: the smallest t at which x -specific reconstruction fails — i.e. the noise level at which the model can no longer recover the specific instance x and must hedge across nearby holons. This marks the *instance-to-class transition*.
- $T_2(x)$: the smallest $t \geq T_1(x)$ at which the reconstruction collapses to the unconditional prior of D . This marks the *class-to-prior transition*.

From these we derive two quantities of interest:

$$\text{Novelty}(x) = T_2(x) - T_1(x), \quad \text{Associativity}(x) = T_2(x).$$

Novelty measures the width of the regime in which x retains class-level identity without being fully specified — the size of the compositional “neighbourhood” it governs. Associativity measures how deeply x is anchored in the data manifold before the reconstruction loses all conditional information.

These two scalars define a coordinate system on the data: every observation maps to a point in the (T_1, T_2) plane. The frontier of known observations is the boundary in this plane beyond which no training data lies, and the estimated location of a new observation can be obtained by:

1. Computing its (T_1, T_2) coordinates from the reconstruction process.
2. Identifying the nearest known holonarchic region in this plane.
3. **Recursively decomposing** the associations active at T_2 — i.e. examining which subsets of D remain influential at the class-to-prior boundary, and applying the same threshold procedure to each subset.

This recursion is the *association deconstruction* step: by repeatedly applying the two-threshold scoring to progressively smaller and more specialized subsets, we descend the holonarchic hierarchy and obtain a multi-resolution characterization of x in terms of compositional dependencies on D . The scoring is not on x in isolation but on *which hierarchical decompositions remain consistent* with x across levels. This is what should be scored: not features, not reconstructions, but the structure of the hierarchy of associations that survive perturbation.

3 A Phenomenological Critique of Smoothness Priors

The dominant regularization practices in deep learning — dropout, L_2 weight decay, input noise — share a common implicit assumption: that the function spaces we wish to learn are *smooth*, in the sense that small perturbations to inputs or weights correspond to small changes in output. This assumption is so deeply embedded that it is often not recognized as an assumption at all.

We argue that this commitment is not metaphysically necessary but reflects the phenomenological structure of human cognition. The way humans perceive and interact with reality — the categories of cause and effect, the expectation of continuous response, the framing of events as smoothly unfolding in time — biases scientific and mathematical formalization toward smooth solution

spaces. The smoothness prior is in this sense an *anthropic projection*: it describes the kinds of regularities that human cognition is equipped to perceive and act upon, not necessarily the structure of the data manifold itself.

This critique opens a question about what alternative priors are available. Sparsity priors (in the Kolmogorov/Occam sense) commit to a different assumption: that good explanations admit compact compositional descriptions, but make no commitment to continuity. Compositional priors — such as the holonarchic structure described in Section 1 — commit to a third position: that good explanations decompose into typed hierarchical associations whose admissible compositions are constrained but not necessarily smooth. The threshold-based scoring of Section 2 is compatible with compositional priors without requiring smoothness.

The practical consequence is that regularization should not be chosen by default but by commitment to a theory of what regularities the system is meant to discover. If the target is compositional generalization across a holonarchic structure, the regularizer should constrain admissible compositions, not penalize discontinuity.

Synthesis

The three insights form a single argument. Holonarchic structure and dispositional K-complexity (Section 1) provide an ontology of what data *is*; threshold-based reconstruction and recursive association decomposition (Section 2) provide an epistemology of how to characterize and place observations within that ontology; and the phenomenological critique (Section 3) clears the ground of inherited regularization commitments that may be incompatible with both. Together they sketch a research program in which the choice of inductive bias is itself a theoretical commitment about the nature of compositional generalization, rather than a hyperparameter to be tuned.

4 Paper Outline (Placeholder)

This appendix sketches the planned structure of the full paper and the experimental program that operationalizes the three insights above. All architectural variants will be reported at a common parameter budget (target: $\approx 50\text{M}$) so that comparisons isolate the effect of inductive bias rather than capacity.

Planned Sections

1. **Introduction.** Motivate compositional generalization as the central problem and position the paper’s contribution as an examination of inductive bias rather than scaling.
2. **Theoretical motivation.** The three insights of this document, expanded with formal definitions of $K(x \mid D, r)$, the two-threshold reconstruction scoring, and the phenomenological critique.
3. **Methods.** The architectural progression and training schemes:
 - (a) Flat baseline transformer (AdamW, no curriculum).
 - (b) Iterative depth-growth training (iter scheme).
 - (c) Iterative + data curriculum (difficulty-ranked exposure).
 - (d) Iterative + curriculum + decaying intra-layer loops with horizon-1 BPTT and PCGrad over loop losses (*loopy iter*).

- (e) Sparsity-prior post-training: iterative magnitude- or activation-based sparsification of the trained dense model, schedule-tuned until the train-validation gap reaches a target.
 - (f) Holonarchic-prior architecture (*diamond*): branch specialization with direction-in-representation-space assignment and radius-as-depth structure, instantiating the holonarchic ontology of Section 1.
4. **Experiments.** Common-budget comparison across all variants on byte-level `enwik8`, reported as bits-per-byte (bpb) on the held-out set, with train-val gap reported alongside.
 5. **Catastrophic-forgetting study.** A targeted experiment to demonstrate the practical value of the holonarchic prior: post-training fine-tuning of a single semantic branch (e.g. a mathematics branch) of the diamond model and measurement of performance preservation on unrelated branches, compared against the same fine-tuning applied to a flat baseline of equal parameter count. The prediction is that the diamond confines fine-tuning updates to the targeted branch and the shared prefix layers, yielding substantially lower degradation on distant capabilities.
 6. **Discussion.** What the comparison reveals about the relative merits of sparsity priors versus compositional/holonarchic priors, and what the catastrophic-forgetting result implies for modular post-training.
 7. **Conclusion.**

Model Variants

Variant	Inductive bias	Status
Flat baseline (AdamW)	none (control)	trained (small)
Iter	depth curriculum	trained
Iter + data curriculum	depth + difficulty curriculum	trained
Iter + curriculum + loops	+ intra-layer recurrence	implemented
Sparsity-pruned (TurboSparse)	sparsity prior	planned
Diamond	holonarchic prior	planned
Diamond + branch fine-tune	forgetting resistance test	planned

Key Empirical Claims to Be Tested

- Iterative depth-growth with data curriculum converges faster per training step than a matched-capacity flat baseline and reaches comparable bpb.
- Sparsity-prior post-training reduces the train-val gap without recourse to smoothness-based regularizers, providing an empirical test of the phenomenological argument of Section 3.
- The diamond architecture achieves bpb competitive with flat baselines at equal parameter count, demonstrating that the holonarchic inductive bias is not prohibitively expensive.
- Branch-localized fine-tuning of the diamond preserves capabilities on distant branches substantially better than equivalent fine-tuning of a flat model. This is the headline result that links the theoretical claims of Sections 1–3 to a measurable practical property.